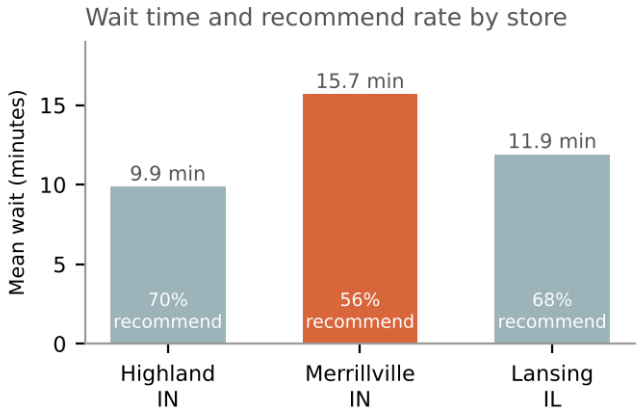
Customer Satisfaction Survey Analysis

Michael Sillyman · Data Analyst portfolio project · Tools: Stata, SPSS · Simulated survey of customers at a 3-store US pharmacy group

Question & method

640 customers at three stores rated their visit. Management wanted to know which parts of the visit drive overall satisfaction, and why the Merrillville store gets fewer recommendations. I ran descriptive statistics, a t-test, a one-way ANOVA with Tukey post hoc comparisons, a chi-square test, Pearson correlations and a multiple regression, scripted in both Stata (.do) and SPSS (.sps) syntax.

Respondents	Avg. satisfaction	Would recommend	Avg. wait	Model R^2
640	**7.6 / 10**	**64%**	**12.6 min**	**.33**





Results

Test	Result	What it means
One-way ANOVA: wait time by store	$F(2, 637) = 142.6$, $p < .001$, $\eta^2 = .31$. Tukey: all three pairs differ ($p < .001$).	Merrillville customers wait 5.8 min longer than Highland and 3.8 min longer than Lansing.
Chi-square: store × recommend	$\chi^2(2) = 11.49$, $p = .003$, Cramér's $V = .13$	Merrillville's recommend rate (56%) is lower than Highland (70%) and Lansing (68%).
Pearson correlation	Wait and satisfaction $r = -.42$; staff help and satisfaction $r = .39$; both $p < .001$	Longer waits go with lower satisfaction; better service goes with higher satisfaction.
t-test: satisfaction, prescription vs other	7.47 vs 7.73; $t(634) = -2.34$, $p = .020$, $d = -.19$	Prescription customers are slightly less satisfied (a small effect).
Multiple regression (N = 621)	$R^2 = .33$, $F(7, 613) = 42.2$, $p < .001$. Wait $b = -0.13$, staff help $b = 0.58$, counseling $b = 0.48$ (all $p < .001$).	Store ($p \geq 0.88$) and visit type ($p = 0.62$) stop mattering once wait time and service are controlled.

What this means

- **Wait time is the strongest single driver.** Each extra minute of waiting lowers satisfaction by 0.13 points, with service and store held constant.
- **Merrillville's problem is its wait, not its location.** Its lower recommend rate disappears in the regression once wait time is accounted for. The same is true of the gap for prescription customers, who simply wait longer.
- **Service quality adds on top:** each point of staff helpfulness adds 0.58 points, and being offered pharmacist counseling adds 0.48.

Recommendations

1. **Cut Merrillville's wait to Highland's level** by adding technician hours at peak times. Closing the 5.8-minute gap would raise satisfaction there by about 0.8 points.
2. **Speed up prescription pickups** with text-when-ready alerts and wider drive-thru use.
3. **Offer counseling at every new prescription** and include helpfulness in staff coaching.

Limitations: the data is simulated and cross-sectional, so the results show association, not proven cause. Satisfaction is self-reported. Regression uses listwise deletion; correlations use pairwise deletion.